

POT: Prototypical Optimal Transport for Weakly Supervised Semantic Segmentation

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Abstract

Weakly Supervised Semantic Segmentation (WSSS) leverages Class Activation Maps (CAMs) to extract spatial information from image-level labels. However, CAMs primarily highlight the most discriminative foreground regions, leading to incomplete results. Prototype-based methods attempt to address this limitation by employing prototype CAMs instead of classifier CAMs. Nevertheless, existing prototype-based methods typically use a single prototype for each class, which is insufficient to capture all attributes of the foreground features due to the significant intra-class variations across different images. Consequently, these methods still struggle with incomplete CAM predictions. In this paper, we propose a novel framework called Prototypical Optimal Transport (POT) for WSSS. POT enhances CAM predictions by dividing features into multiple clusters and activating each cluster using its prototype. In this process, a similarity-aware optimal transport is employed to assign features to the most probable clusters. This similarity-aware strategy ensures the prioritization of significant cluster prototypes, thereby improving the accuracy of feature assignment. Additionally, we introduce an adaptive OT-based consistency loss to refine feature representations. This framework effectively overcomes the limitations of single-prototype methods, providing more complete and accurate CAM predictions. Extensive experimental results on standard WSSS benchmarks (PASCAL VOC and MS COCO) demonstrate that our method significantly improves the quality of CAMs and achieves state-of-the-art performances. The source code will be released <https://github.com/jianwang91/POT>.

1. Introduction

Weakly supervised semantic segmentation (WSSS) with image-level labels reduces the labor for pixel-level annotations, thereby receiving increasing attention. In this task, Class Activation Maps (CAMs) [59] are widely used to

localize objects and employed to generate pseudo-labels, which are further utilized to train a segmentation model [8]. However, CAMs primarily highlight the most distinctive foreground regions [36, 53], providing insufficient supervision for training the segmentation model [8].

To mitigate this limitation, prototype-based methods [9, 57, 58] are introduced. These methods first generate binary masks for each class, either by feature similarity [9, 57] or a pretrained segmentation model [58]. These binary masks are then used to perform mask average pooling on the corresponding features to construct a prototype for each class within an image. The resulting prototypes capture the essential characteristics of foreground classes and are leveraged to activate features using cosine similarity. Compared to classifier CAMs, these prototype CAMs are more complete, thereby enhancing the overall performance.

However, prototype CAMs still fail to cover complete regions of foreground objects in certain images. Feature distribution of such images reveals that the foreground object features are often sparse and dispersed (see Fig. 1 (a)). This issue is primarily attributed to the significant intra-class variations [37] in the dataset, which limits the ability of a single prototype to activate all relevant features. Specifically, the classifier aims to capture global discriminative intra-class characteristics to accurately classify an image. Features that encapsulate such characteristics (e.g., faces within the person class) can be effectively recognized and grouped to form a compact primary cluster. Nevertheless, regions belonging to the same class may exhibit highly diverse appearances, even within the same image (see Fig. 1 (a), areas in the red and green boxes). These image-specific features cannot be closely clustered into the previously described primary cluster due to their lack of global intra-class characteristics. Instead, these feature clusters often appear separately from the primary one. When constructing prototype CAMs with a single prototype, only features in close proximity to the prototype are fully activated (red area in Fig. 1 (a)), while other features are only partially activated (green area in Fig. 1 (a)). This incomplete activation fails to provide sufficient guidance for segmentation model training, leading to unsatisfactory performance.

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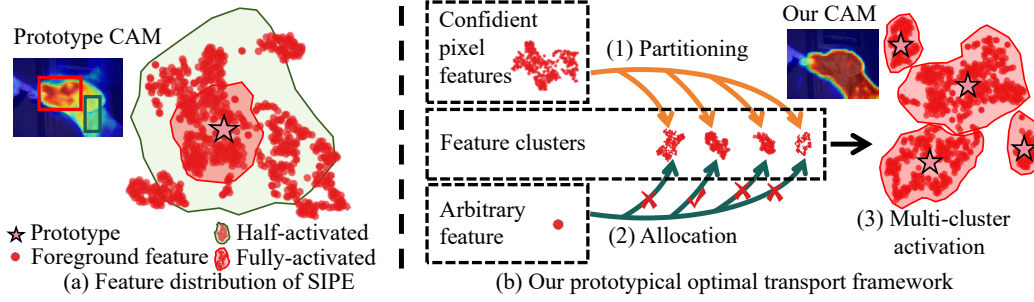


Figure 1. (a): Feature distribution of single-prototype methods (e.g., SIPE [9]): some clusters are positioned at a considerable distance from the cluster containing the class prototype. Only features in close proximity to the prototype can be fully activated, leading to incomplete prototype CAMs. Fully-activated: feature activated with high prediction score (e.g., score > 0.6). Half-activated: feature activated with low prediction score (e.g., score < 0.4). (b): Our framework divides features into multiple clusters and activates the features separately.

To address the aforementioned challenge, we propose a novel plug-and-play framework called **Prototypical Optimal Transport (POT)** for WSSS, as shown in Fig. 1 (b). Considering the limitation of a single prototype in capturing the diverse attributes of foreground features, POT divides feature vectors into multiple clusters and generates multiple cluster prototypes to encapsulate a broader range of attributes within an image. By combining the activation results from each cluster, POT obtains comprehensive foreground CAM results. Specifically, feature vectors with confident CAM predictions are initially divided into multiple clusters. Each cluster encapsulates the most distinctive characteristics of a class and serves as an anchor for feature allocation within the image. Subsequently, a prototypical optimal transport allocation is performed to assign each feature to the most appropriate cluster. Once the allocation is completed, features are activated by their corresponding cluster prototypes using cosine similarity. Finally, the cluster-activated results of the same class are combined to generate complete foreground CAM predictions.

In the above pipeline, accurately assigning features to their corresponding clusters is both crucial and challenging. The use of multiple clusters per class further increases the complexity of the allocation process. Incorrectly assigned features cannot be fully activated, resulting in incomplete CAM predictions. To address this challenge, optimal transport (OT) theory [39] is employed to learn an adaptive transport plan for assigning features to their potential clusters effectively. However, conventional OT employs an average constraint on target distribution, which treats all cluster prototypes equally. This strategy overlooks the varying significance and importance of different cluster prototypes, potentially leading to suboptimal allocation. In contrast, we introduce a similarity-aware constraint to address this limitation. This constraint prioritizes cluster prototypes with high similarities to the classifier CAM prototypes, recognizing them as more critical. This bias allows the OT optimization process to focus more on the prototypes containing

discriminative characteristics when determining the lowest cost through the matching plan. Based on the transport plan, an adaptive OT-based consistency loss is further proposed to optimize the feature representation in POT. Our contributions can be summarized as follows:

- The reason for sparse and dispersed feature distribution in current prototype-based methods is analyzed, revealing that a single prototype is insufficient to capture all attributes of foreground features. This limitation leads to incomplete foreground CAM predictions.
- A novel plug-and-play prototypical optimal transport framework for WSSS is proposed. This framework employs optimal transport to effectively assign features to multiple clusters. By combining cluster-activated results of the same class, the framework enhances CAM predictions on various existing WSSS methods.
- Our method achieves new state-of-the-art benchmarks in multistage WSSS, attaining 76.1% and 76.7% mIoU on the validation and test sets of Pascal VOC 2012 and 47.9% mIoU on the MS COCO 2014 validation set.

2. Related Work

Weakly Supervised Semantic Segmentation. Class Activation Maps (CAMs) [59] are extensively utilized in WSSS with image-level labels. However, CAMs primarily focus on the most discriminative regions of images. To address this limitation, a variety of methods explore diverse strategies, including attention mapping [38, 40, 42, 52], erasing and accumulation [21, 22, 51], region growing [19, 55], boundary constraints [6, 27, 33], cross-image information mining [41, 44, 48], and prototype-based methods. Among them, prototype-based methods [9, 10, 15, 37, 49] have drawn increasing attention due to their effectiveness in enhancing CAM completeness. Chen et al. [9] introduce image-specific prototypes by evaluating class-wise Intersection over Union (IoU) between the structure map and classifier CAMs. Du et al. [15] propose a pixel-to-prototype contrast method, which involves selecting top-K pixel fea-

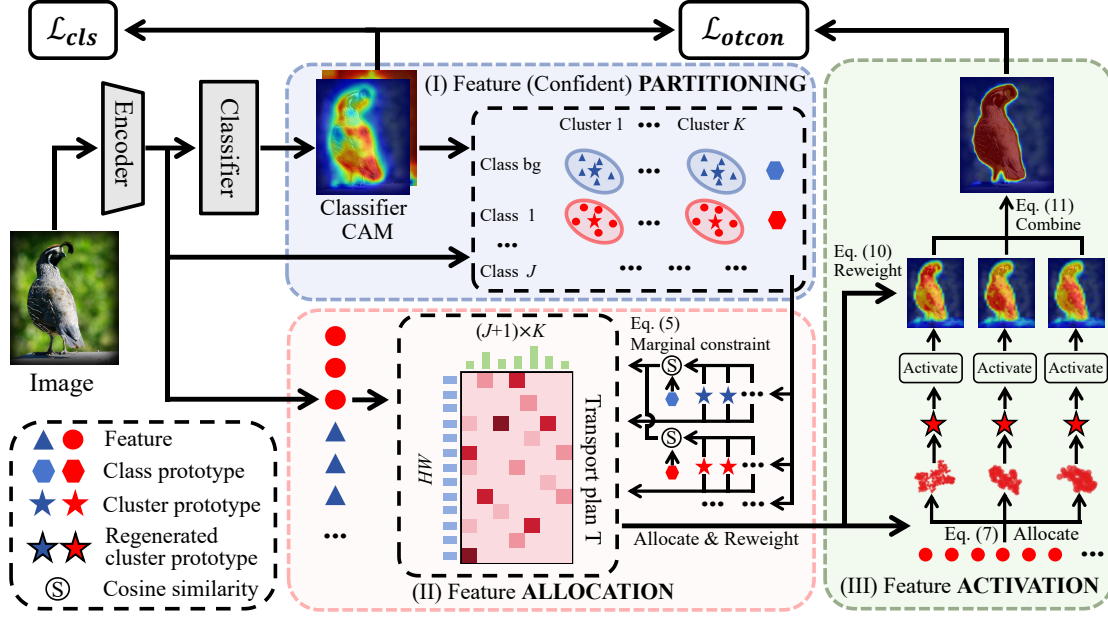


Figure 2. Illustration of POT. The framework begins by partitioning features with confident CAM values into multiple clusters. Then, an optimal transport is constructed to allocate all features to these clusters. This allocation is guided by a marginal constraint, which uses cosine similarities between cluster prototypes and class prototypes to ensure accurate allocation. Once the features are allocated, new prototypes are generated using all the features within each cluster in feature activation stage. These newly generated prototypes are used to activate their corresponding feature clusters. The activated results are reweighted by the transport plan and combined to form the final CAM predictions. Additionally, an OT-based consistency loss is applied between the CAM predictions and the classifier CAMs to optimize feature representations within the framework. For simplicity, the process in feature activation stage is demonstrated with a single class.

tures with confident CAM values to compute prototypes. Tang et al. [37] refine prototypes by aligning feature distributions between context prototypes over the whole dataset and instance prototypes of the current image. The generated prototypes are further used to construct prototype CAMs based on cosine similarity between them and feature vectors. Despite their effectiveness, these prototype-construct strategies generally rely on a single prototype per class to activate features, which is inadequate given the huge intra-class variations in the dataset. Unlike these approaches, our method leverages multiple cluster prototypes to generate prototype CAMs, providing a more comprehensive representation of foreground classes. While Chen et al. [10] utilize multiple clusters as well, their approach generates such clusters across the dataset, relying on manually set thresholds to select prototypes. This operation limits their performance. In contrast, our method employs optimal transport to learn an adaptive allocation strategy, leading to improved results without the need for manual parameter tuning.

Optimal Transport. Optimal Transport (OT) [39] offers an effective method for transferring mass between two distributions. To reduce the computational cost of OT, Sinkhorn OT [14, 23] is proposed, which can be efficiently solved using the Sinkhorn algorithm. This algorithm ensures the widespread applicability of OT among various fields. In

computer vision, OT has been successfully applied to many different tasks, such as generative models [2, 56], structural matching [45, 56], image matching [54], and domain adaptation [3, 4, 28, 50]. All these applications seek to find an optimal plan that minimizes a specific cost function. In our work, we employ optimal transport to create an allocation plan that assigns features to the most appropriate clusters. By learning such an adaptive transport plan [34], we improve the efficiency and effectiveness of feature distribution, allowing for more precise feature clustering.

3. Methodology

This section presents a detailed explanation of the proposed Prototypical Optimal Transport (POT) framework for WSSS, as depicted in Fig. 2. The framework includes three key stages: feature partitioning, feature allocation, and feature activation. In the first stage, feature vectors with confident classifier CAM values are grouped into a fixed number of clusters for each class. These clusters capture the most critical characteristics of their respective classes, and a cluster prototype is generated for each of them to serve as an anchor for the subsequent feature allocation. The second stage employs Optimal Transport (OT) to allocate features to clusters. The resulting transport plan provides the probability distribution for assigning each feature to a specific

cluster prototype established in the first stage, thereby guiding the systematic allocation of feature vectors. In the final stage, the cluster prototypes are updated based on the newly allocated feature vectors, and features are activated individually by calculating cosine similarities with their corresponding cluster prototypes. In addition, an OT-based consistency loss is introduced to maintain the consistency of the final activation results and the classifier CAMs, which optimizes the feature representation and provides an effective and bounded exploration of prototypes.

3.1. Feature Partitioning

This section provides a detailed description of the feature partitioning stage. For an input image, $\mathbf{F}_4 \in \mathbb{R}^{D \times H \times W}$ represents the feature map produced by the last layer of the encoder, where D , H , and W denotes the dimensions of the feature map. Additionally, the classifier weight for each class c is denoted as $\theta_c \in \mathbb{R}^{D \times 1}$. The original classifier CAMs are derived as follows:

$$\mathcal{M}^{\text{cls}}(\mathbf{F}_4, \theta) = \underbrace{\left\{ \text{ReLU}(\theta_1^T \mathbf{F}_4), \dots, \text{ReLU}(\theta_{|\mathcal{C}|}^T \mathbf{F}_4) \right\}}_{|\mathcal{C}| \text{ foreground classes}} \cup \underbrace{\left\{ (1 - \max_{c \in \mathcal{C}} (\text{ReLU}(\theta_c^T \mathbf{F}_4))) \right\}}_{1 \text{ background class}}, \quad (1)$$

where T denotes the transpose operation and $|\mathcal{C}|$ denotes the number of foreground classes in set $\mathcal{C}$.

The classifier CAMs effectively highlight the most discriminative regions for each category [10, 42]. To capture the significant representations of these regions for the subsequent feature allocation stage, CAM-based prototypes are established as anchors for each class in an image. However, previous methods [9, 57, 58] typically rely on a single prototype per class, which often fails to capture the full range of attributes within each foreground class due to their substantial intra-class variations [37]. Therefore, they can only activate features in close proximity to the prototype, leading to incomplete foreground CAM predictions. To address this limitation, we propose the use of multiple prototypes within each class. Each of these cluster prototypes captures attributes from features within its specific cluster, enabling a more comprehensive activation of features and a more accurate representation of the class. Specifically, the k -means approach is used to separate features in activated regions for each class. Features with CAM values above a threshold τ are clustered into K groups for each class (including background class) present in the image. The cluster prototypes are represented as $\mathbf{P} \in \mathbb{R}^{(J+1) \times K \times D}$, where J denotes the number of foreground classes and $\mathbf{P}_{c,k}$ denotes the prototype for the k -th cluster of class c . The cluster prototypes

can be defined as:

$$\mathbf{P} = \text{MAP}(\text{Mask}(\mathbf{F}_4, \mathcal{M}^{\text{cls}}) \odot L(\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4)), \quad (2)$$

where $\text{MAP}(\cdot)$ denotes average pooling over the spatial dimension of flattened features. $\text{Mask}(\cdot)$ generates a binary mask $\text{Mask} \in \mathbb{R}^{HW \times (J+1) \times K}$ according to the clustering results of confident features, where $\text{Mask}_{i,c,k} = 1$ indicates the feature at position i is clustered to the k -th cluster of class c . $\mathbf{F}_i$ represents the feature map produced by the i -th layer of encoder. The function $L(\cdot)$ represents a linear projection, and $\odot$ denotes element-wise multiplication.

3.2. Feature Allocation

Following feature partitioning in Section 3.1, the next stage involves allocating all the feature vectors to the previously generated clusters. The key challenge in this stage is to effectively measure the distances between features and candidate prototypes. Common distance metrics, such as cosine distance, typically focus on point-to-point solutions. These methods calculate the distance between each feature and prototype and then select the closest prototype for feature allocation. However, these methods ignore the interactions among features and treat each allocation individually. In practice, feature allocation in this task should be considered a set-to-set problem, where the allocation of certain features affects the subsequent allocation of others. To address such a problem, Optimal Transport (OT) theory provides a comprehensive framework. We first briefly introduce OT theory and then demonstrate its role in the feature allocation stage. Essentially, OT aims to optimize the transfer of mass between two distributions under specified marginal constraints [39]. Let M and N denote the lengths of these two distributions. The cost matrix associated with the transfer between these two distributions is represented as $\mathbf{C} \in \mathbb{R}^{M \times N}$. The OT problem can be defined as:

$$\begin{aligned} \mathbf{T}^*(\mathbf{C}) &= \arg \min_{\mathbf{T}} \sum_{i=1}^M \sum_{j=1}^N \mathbf{T}_{i,j} \mathbf{C}_{i,j}, \\ \text{s. t. } \mathbf{T} \mathbf{1}_N &= \mathbf{u}, \quad \mathbf{T}^T \mathbf{1}_M = \mathbf{v}, \end{aligned} \quad (3)$$

where $\mathbf{T}$ is the transport plan matrix. $\mathbf{1}_M$ and $\mathbf{1}_N$ are column vectors of ones with dimensions M and N , respectively. $\mathbf{u}$ and $\mathbf{v}$ are the source and target probability distributions. Typically, these distributions are set to uniform distributions [28, 35], i.e., $\mathbf{u}_i = 1/M$ and $\mathbf{v}_j = 1/N$, implying that the mass in the source distribution is equally transported to the target distribution.

In our scenario, features and prototypes are modeled as two distributions, and the ultimate goal is to assign features to corresponding prototypes based on their pairwise distances. The optimal transport theory allows us to find the best strategy to assign these features, minimizing the total

distance across all possible assignments. However, traditional OT methods treat all cluster prototypes equally and ignore the varying importance of these cluster prototypes in each class. To address this limitation, we introduce a similarity-aware optimal transport method. The similarities between the cluster prototypes and corresponding classifier CAM prototypes are leveraged to generate a similarity-aware marginal constraint for optimal transport. Specifically, a similarity matrix, denoted as $\mathbf{X} \in \mathbb{R}^{(J+1) \times K}$, is constructed where each element $x_{c,k}$ represents the similarity between the k -th cluster prototype $\mathbf{P}_{c,k}$ of class c and the classifier CAM prototype $\mathbf{P}_c^{\text{cls}}$ of the same class c . This similarity is defined as follows:

$$x_{c,k} = \text{sim}(\mathbf{P}_{c,k}, \mathbf{P}_c^{\text{cls}}), \quad (4)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity. The matrix $\mathbf{X}$ is first flattened and then passed through a softmax function to construct the marginal similarity-aware constraint:

$$\mathbf{x} = \text{softmax}(\text{Flat}(\mathbf{X})), \quad (5)$$

where $\text{Flat}(\cdot)$ denotes flattening operation. This transformation ensures that the marginal constraints are normalized and represent the relative importance of cluster prototypes.

Marginal constraints are used to define the distributions in optimal transport. In previous methods [4, 35], average constraint is applied to the target distribution, which results in a uniform allocation of features across all cluster prototypes. Instead, we replace the average constraint with the similarity-aware constraint from Eq. (5) on the target distribution. This modification enables the transport process to prioritize cluster prototypes with higher similarities to the classifier prototype, thereby allocating more features to high-similarity cluster prototypes while reducing allocation to those with lower similarities. To find the optimal transport plan $\mathbf{T} \in \mathbb{R}^{M \times N}$ that maximizes the similarities (i.e., minimizes the pairwise distances) between feature vectors and cluster prototypes under our proposed similarity-aware constraint, the objective function is formulated as follows:

$$\begin{aligned} \mathbf{T}^*(\mathbf{S}) &= \arg \max_{\mathbf{T}} \sum_{i=1}^M \sum_{j=1}^N \mathbf{T}_{i,j} \mathbf{S}_{i,j}, \\ \text{s. t. } \mathbf{T} \mathbf{1}_N &= \frac{1}{M} \mathbf{1}_M, \quad \mathbf{T}^T \mathbf{1}_M = \mathbf{x}, \end{aligned} \quad (6)$$

where $\mathbf{S} \in \mathbb{R}^{M \times N}$ is a similarity map and the element $\mathbf{S}_{i,j}$ represents the cosine similarity between the i -th feature and the j -th cluster prototype. N is the total number of prototypes and $N = (J+1) \times K$. M is the number of feature vectors and $M = HW$. $\mathbf{1}_M$ is a column vector of ones with dimension M . To improve the computational efficiency, Sinkhorn distance [13] is employed for fast optimization as in previous methods [4, 5].

Upon completion of the optimization process, we obtain an optimized transport plan $\mathbf{T}^*$. The elements in the i -th row of $\mathbf{T}^*$ represent a categorical distribution, indicating the likelihood of allocating the i -th feature to each cluster prototype. Finally, following a greedy strategy, the i -th feature is assigned to the prototype with the highest probability, as in the previous methods [4, 35]:

$$j^* = \arg \max_j \mathbf{T}_{i,j}^*. \quad (7)$$

Through the optimized transport plan, each feature is assigned to a specific cluster prototype. This assignment process promotes the clustering of similar features in the feature space, thereby enhancing the representation capability of the model.

3.3. Feature Activation

The optimal transport plan elements indicate the likelihood of allocating feature vectors to various prototypes, and they are employed to reweight the prototype CAMs and weight the training loss in our POT framework.

Reweighting Prototype CAMs. The initial cluster prototypes in Eq. (2) are obtained from confident CAM predictions. To fully activate features within each cluster, it is essential to update these cluster prototypes by incorporating newly assigned features according to the transport plan allocation in the preceding stage. Specifically, cluster prototypes $\mathbf{P}^* \in \mathbb{R}^{(J+1) \times K \times D}$ are re-generated using all features allocated to them:

$$\mathbf{P}^* = \text{MAP}(\text{Mask}^*(\mathbf{T}^*) \odot L(\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4)). \quad (8)$$

The function $\text{Mask}^*(\cdot)$ generates a re-allocation binary mask $\text{Mask}^* \in \mathbb{R}^{HW \times (J+1) \times K}$, where an element $\text{Mask}_{i,c,k}^* = 1$ indicates that the feature at position i is assigned to the k -th cluster of class c according to the transport plan. $\mathbf{F}_i$ represents the feature map produced by the i -th layer of the encoder. By incorporating all allocated features, the updated prototypes encapsulate a more comprehensive representation within each cluster, including both confident and less confident features. This operation enables the prototypes to activate features within their respective clusters more effectively compared to those derived solely from confident CAM values in Eq. (2).

Subsequently, the prototype CAM for the k -th cluster of class c , denoted as $\mathcal{M}_{c,k}^{\text{p}}$, is derived by calculating the cosine similarity between the hierarchical features and its updated prototype $\mathbf{P}_{c,k}^*$ as follows:

$$\mathcal{M}_{c,k}^{\text{p}} = \text{ReLU}(\text{sim}(\mathbf{P}_{c,k}^*, L(\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4))). \quad (9)$$

The final prototype CAMs are constructed by aggregating the pixel-weighted results from the obtained prototype CAMs. Initially, these prototype CAMs are reweighted using the optimized transport plan. The elements allocated to

the k -th cluster of class c in the transport plan $\mathbf{T}^*$, denoted as $\mathbf{T}_{c,k}^*$, are used to reweight the corresponding prototype CAMs. This operation prioritizes feature-prototype pairs with higher similarities while de-emphasizing those with minimal similarities, ensuring a more accurate representation of the relationships between features and prototypes:

$$\mathcal{M}_{c,k}^{p*} = \mathbf{T}_{c,k}^* \odot \mathcal{M}_{c,k}^p. \quad (10)$$

The refined prototype CAMs contain K activated results for each class. Each of these results partially activates a distinct region within that class. To construct the final prototype CAM for each class c , these K activation results are aggregated by summing them together:

$$\mathcal{M}_c^{p*} = \sum_{i=1}^K \mathcal{M}_{c,k}^{p*}. \quad (11)$$

Weighting Consistency Loss. To ensure that the generated prototype CAM does not diverge significantly from the classifier CAM, we introduce an OT-based consistency loss, which leverages the transport plan derived in Eq. (6). Unlike the uniform-based consistency loss in previous methods [9, 57, 58], which disregards the certainty in the predictions and treats all pixels equally, our OT-based consistency loss employs a pixel-level adaptive weights. This weight adjusts the contribution of each pixel in the loss computation between the optimized prototype CAMs and the classifier CAMs. It enforces strong constraints on pixels with high certainty in their predictions while relieving them on those with lower certainty (i.e., the prediction value closes to 0.5). Specifically, the adaptive weight is applied along the class dimension for each pixel, and the OT-based consistency loss is defined as follows:

$$\mathcal{L}_{otcon} = \frac{1}{HW} \sum_{i=1}^{HW} \sum_{c=1}^{|\mathcal{C}|+1} w_{i,c} \|\mathcal{M}_{i,c}^{p*} - \mathcal{M}_{i,c}^{cls}\|_1, \quad (12)$$

where $\|\cdot\|$ denotes the ℓ_1 -normalization. $\mathcal{M}^{p*}$ and $\mathcal{M}^{cls}$ denote the optimized prototype CAMs and the classifier CAMs. The coefficient $w_{i,c}$ is derived from the transport plan for each foreground class c as well as the background:

$$w_{i,c} = l_c \left(1 - \frac{1}{\log(|\mathcal{C}|+1)} \mathcal{H}(g(t_{i,c})) \right), \quad (13)$$

where l_c represents the one-hot label for the c -th class, $1/\log(|\mathcal{C}|+1)$ is a scaling factor and $g(t_{i,c}) = \sum_{k=1}^K t_{i,c,k}$. The $t_{i,c,k}$ denotes the value in the i -th row allocated to the k -th cluster of class c within the transport plan. The sum of the K values for a specific feature at position i and class c , referred to as $g(t_{i,c})$, indicates the probability of the feature belonging to this class [4, 5]. The binary entropy $\mathcal{H}(g(t_{i,c}))$ of the distribution $g(t_{i,c})$ is defined as:

$$\mathcal{H}(g(t_{i,c})) = -g(t_{i,c}) \log g(t_{i,c}) - (1-g(t_{i,c})) \log(1-g(t_{i,c})). \quad (14)$$

Method	PASCAL VOC	
	CAM	Mask
AdvCAM _{CVPR'21} [24]	55.6	69.9
CLIMS _{CVPR'22} [43]	56.6	70.5
SIPE _{CVPR'22} [9]	58.6	68.0
AEFT _{ECCV'22} [51]	56.0	71.0
MCTformer _{CVPR'22} [46]	61.7	69.1
AMN _{CVPR'22} [26]	62.1	72.2
LPCAM _{CVPR'23} [10]	65.3	72.7
MCTformer+ _{TPAMI'24} [47]	68.8	76.2
CLIP-ES+CPAL _{CVPR'24} [37]	71.9	75.8
CLIP-ES+PSDPM _{CVPR'24} [58]	-	77.3
SFC _{AAAI'24} [57]	64.7	73.7
SFC + Ours	68.5	76.5
CLIP-ES _{CVPR'23} [31]	70.8	75.0
CLIP-ES + Ours	75.0	79.3

Table 1. Evaluation (mIoU (%)) of CAM and pseudo-label (denoted as Mask) results on PASCAL VOC 2012 train set.

The value of $\mathcal{H}(g(t_{i,c}))$ approaches the maximum as uncertainty increases and conversely approaches 0 when uncertainty is low. As a result, the adaptive coefficient $w_{i,c}$ dynamically adjusts the contribution from each pixel, assigning lower weights to pixels with higher prediction uncertainty. By incorporating this adaptive weighting scheme, our method enforces varying constraint levels between optimized prototype CAMs and classifier CAMs, enhancing the efficiency and effectiveness of the training process.

3.4. Overall Training Objective

We utilize the multi-label classification loss $\mathcal{L}_{cls}$, together with the OT-based consistency loss $\mathcal{L}_{otcon}$, to train our model. The overall loss function is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \lambda \mathcal{L}_{otcon}, \quad (15)$$

where λ is the weighting parameter. The multi-label soft margin classification loss $\mathcal{L}_{cls}$ is expressed as:

$$\mathcal{L}_{cls} = \frac{1}{|\mathcal{C}|} \sum_{c=1}^{|\mathcal{C}|} l_c \log(\text{sigmoid}(GAP(\theta_c^T F))) + (1-l_c) \log(1 - \text{sigmoid}(GAP(\theta_c^T F))), \quad (16)$$

where $GAP(\cdot)$ function denotes the global average pooling.

4. Experiment

4.1. Experiment Settings

Datasets and Evaluation Metrics. Our approach is evaluated on two widely used benchmark datasets: PASCAL VOC 2012 [16] and MS COCO 2014 [30]. PASCAL VOC

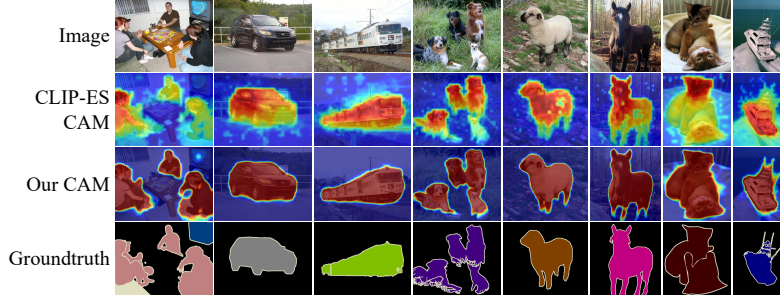


Figure 3. Visualization results on PASCAL VOC 2012 dataset. Our approach generates more comprehensive visual results compared to the results from CLIP-ES [31].

2012 contains 20 foreground classes, and its extended training set, SBD [17], which includes 10,582 images, is used for training. MS COCO 2014 consists of 80 foreground classes, with 80k images for training and 40k images for validation. The mean Intersection-over-Union (mIoU) [32] is used as the evaluation metric to assess the performance of the models across all experiments.

Implementation Details. For pseudo-label generation, we adopt the ImageNet pre-trained ResNet50 [18]. Training is conducted with the stochastic gradient descent (SGD) optimizer for 6 epochs with a batch size of 16. The learning rates for the Encoder and additional layers are set to 0.1 and 1, respectively, following prior research [9, 43]. A polynomial learning rate scheduler with a power of 0.9 is also implemented. The loss coefficient λ is set to 3 in Eq. (15). The parameter K for k-means clustering and the threshold τ in Section 3.1 are set to 3 and 0.7, respectively. Pseudo-labels are generated leveraging DenseCRF [20] and IRN [1] following previous methods [9, 57, 58].

4.2. Comparison with the State-of-the-art

Comparison of Pseudo-label Quality. To evaluate the effectiveness of our approach, we conduct a comparative analysis focusing on CAMs (denoted as CAM) and pseudo masks (denoted as Mask), as outlined in Tab. 1. Our method is incorporated into two prominent baseline models: CLIP-ES [31] and SFC [57]. The experimental results in Tab. 1 demonstrate the consistent improvements across both baseline methods. Notably, when applied to the CLIP-ES [31] baseline, our method increases CAM and pseudo-label quality by 4.2% and 4.3%, respectively, surpassing all other WSSS methodologies. To illustrate these improvements, we compare the qualitative outcomes of CAMs generated by our method against those from CLIP-ES [31]. As depicted in Fig. 3, our approach consistently delivers more comprehensive results than the baseline method.

Comparison of WSSS Performance. In WSSS, pseudo masks are employed as groundtruth to fully supervise the training of segmentation models [9, 43]. Tab. 2 presents the quantitative comparison of our method and recent state-of-

Method	VOC		MS COCO
	val	test	val
AdvCAM _{CVPR'21} [24]	68.1	68.0	42.6
ReCAM _{CVPR'22} [11]	68.5	68.4	-
URN _{AAAI'22} [29]	69.5	69.7	40.7
SIPE _{CVPR'22} [9]	68.8	69.7	43.6
AMN _{CVPR'22} [26]	69.5	69.6	44.7
W-OoD _{CVPR'22} [25]	70.7	70.1	-
AEFT _{ECCV'22} [51]	70.9	71.7	44.8
FPR _{ICCV'23} [7]	70.0	70.6	43.9
OCR+SEAM _{CVPR'23} [12]	67.8	68.4	-
LPCAM _{CVPR'23} [10]	70.1	70.4	45.5
FPR _{ICCV'23} [7]	70.3	70.1	-
CLIP-ES+PSDPM _{CVPR'24} [58]	74.1	74.9	47.2
CLIP-ES+CPAL _{CVPR'24} [37]	74.5	74.7	46.8
SFC _{AAAI'24} [57]	71.2	72.5	46.8
SFC + Ours	74.5	75.2	47.6
CLIP-ES _{CVPR'23} [31]	71.1	71.4	45.4
CLIP-ES + Ours	76.1	76.7	47.9

Table 2. Evaluation (mIoU (%)) of segmentation performances on PASCAL VOC 2012 and MS COCO 2014 datasets.

the-art models on the PASCAL VOC 2012 and MS COCO 2014 datasets. Our method achieves 76.1% and 76.7% mIoUs using an ImageNet pre-trained backbone on PASCAL VOC 2012 and 47.9% mIoU on MS COCO 2014, surpassing all other multistage image-level WSSS methods. Furthermore, the consistent performance improvements achieved by our framework across various baseline methods and datasets highlight that these original single-prototype baselines are limited by their single-prototype design. Our approach can effectively alleviate this limitation and achieve substantial performance improvement.

4.3. Ablation Studies

To thoroughly assess the effectiveness of the proposed method, we conduct ablation studies focusing on three key components: multi-cluster activation, similarity-aware OT, and OT-based consistency loss. The results are presented in Tab. 3 and Tab. 4. In these experiments, we integrate our method into the CLIP-ES model [31]. Since CLIP-

#	Base	Single-cluster	Multi-cluster	SOT & $\mathcal{L}_{otcon}$	mIoU (%)
I	✓				70.8
II	✓	✓			71.5
III	✓		✓		72.9
IV	✓		✓	✓	75.0

Table 3. Ablation study of our framework on PASCAL VOC 2012 train set. The mIoU (%) metric is used for evaluation. CLIP-ES [31] is adopted as the baseline method. ‘SOT’ denotes similarity-aware optimal transport.

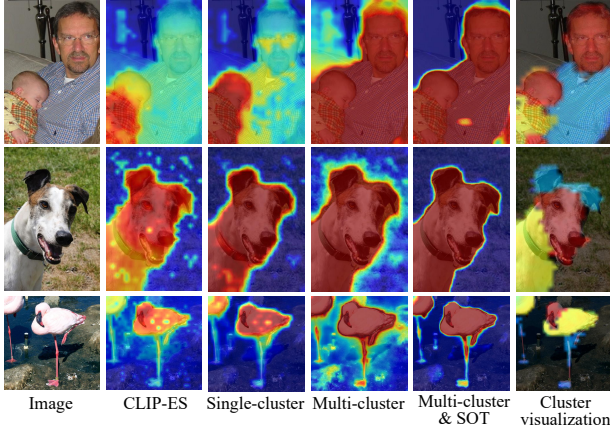


Figure 4. Visual comparison of important components.

ES is a training-free model, we distill its knowledge into a ResNet-50 encoder, using a single-cluster activation setup as the baseline for comparison. Detailed structural information can be found in the supplementary material.

Multi-cluster Activation. We conduct ablation studies in conditions II and III to evaluate the effectiveness of multi-cluster activation. To avoid the influence of OT, we conduct these two experiments without using OT. Instead, we directly allocate features to specific classes based on the cosine similarity with class prototypes. After that, prototypes are regenerated and features are activated in line with the POT pipeline. By comparing conditions II and III, we find that multi-cluster activation results in an improvement of 1.4% in performance. Visualizations of sample results, shown in Fig. 4 (3rd and 4th columns), illustrate that multi-cluster activation leads to more comprehensive CAM predictions. Both the quantitative and qualitative results highlight the advantages and effectiveness of multi-cluster activation in our framework.

Similarity-aware OT. Tab. 4 presents a detailed analysis of the improvements achieved by OT, using condition III as the baseline for comparison. Applying vanilla OT leads to a 1.2% increase in mIoU, achieving a performance of 74.1%. Further incorporating similarity-aware OT results in an additional 0.5% performance boost, demonstrating the advantages of the similarity-aware constraint in the optimal

#	Base	OT	SOT	$\mathcal{L}_{con}$	$\mathcal{L}_{otcon}$	mIoU (%)
III	✓			✓		72.9
V	✓	✓		✓		74.1
VI	✓		✓	✓		74.6
VII	✓		✓		✓	75.0

Table 4. Ablation study of OT-enhanced components on PASCAL VOC 2012 train set. The mIoU (%) metric is used for evaluation. Condition III is adopted as the baseline in this table. ‘ $\mathcal{L}_{con}$ ’ refers to the loss calculated without including the coefficient $w_{i,c}$.

transport framework. Visual comparisons between results obtained with similarity-aware OT (SOT) and those without OT (the 5th and 4th columns in Fig. 4, respectively) show that integrating SOT yields more complete and less noisy foreground CAM predictions. One explanation is that directly using cosine similarity for feature allocation can mistakenly incorporate background features into the foreground pixel clusters. Consequently, prototypes generated from these mixed clusters may inadvertently activate background features. In contrast, SOT provides more accurate feature allocation, as shown in the 6th column in Fig. 4, which visualizes three clusters of the foreground class. This approach results in cleaner and more distinct prototypes. Furthermore, the reweighting operation in Eq. (10) also diminishes the background noise in CAM predictions.

OT-based Consistency Loss. When comparing experiments VI and VII in Tab. 4, it is clear that using OT-based consistency loss can enhance CAM performance, achieving a final mIoU of 75.0. This result underscores the accuracy and effectiveness of the optimized transport plan.

5. Conclusion

This paper introduces the POT framework, a novel approach designed to address the issue of incomplete CAM predictions in existing prototype-based approaches. We demonstrate that a single prototype is insufficient to capture the diverse attributes of foreground objects due to intra-class variations, leading to incomplete CAMs. By utilizing multiple feature clusters and a similarity-aware optimal transport strategy, POT achieves precise feature allocation, obtaining comprehensive and accurate results. Additionally, we incorporate an OT-based consistency loss to optimize feature representations and constrain the prototype exploration, further enhancing the effectiveness of the framework. Extensive experiments on two popular datasets demonstrate the superiority of our proposed method in image-level WSSS.

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